

# GO-EUM CHA

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## RESEARCH INTERESTS

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I study how socially interactive robots and conversational AI can support people in emotionally and cognitively demanding contexts, such as mental health, learning, and caregiving, through multimodal behavioral and interactional signals. Current work examines how these signals shape engagement and communication in human-robot and human-AI interaction.

## EDUCATION

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**Purdue University** *2025 – Present*  
Ph.D. Student, Computer Science (formerly affiliated Technology, 2021–2024)

**Purdue University** *2018 – 2020*  
Master of Science, Computer and Information Technology  
Thesis: Cognitive Load Estimation with Behavioral Cues in Human-Machine Interaction.

**Soongsil University** *2011 – 2017*  
Bachelor of Science, College of Computer Science & Engineering  
Bachelor of Engineering, College of Global School of Media

## PROFESSIONAL EXPERIENCE

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**Graduate Research Assistant** *2025 – Present*  
Computer Science, Purdue University  
- Research on conversational AI, multimodal sensing, and human-agent interaction.

Computer Information and Technology, Purdue University *Jan. 2020 – May 2020*  
- Analyzed student retention and admission-related data for a departmental retention project.

Computer Information and Technology, Purdue University *Aug. 2018 – May 2019*  
- Developed autonomous vehicle control systems using beacon sensors and ROS in a leader-follower setting.

**Research Aide**  
Mathematics and Computer Science Division, Argonne National Laboratory *May 2019 – Aug. 2019*  
- Developed computer vision models for cloud coverage detection using statistical segmentation, SVMs, and CNN U-Net architectures.  
- Developed machine learning models for deployment on Waggle sensor platforms.

**iOS Software Engineer → Software Engineering Manager** *2015 – 2018*  
Everysing Korea Co., Ltd., iTech Engineering Co. Ltd.  
- Built iOS applications, collaborating with cross-functional teams.  
- Led engineering efforts on reliability analysis and learning systems, including training on power plant simulators.

## TEACHING EXPERIENCE

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### Teaching Assistant, Purdue University, IN, USA

- CS 47500 Human-Computer Interaction Spring 2026  
Concepts and Research in Human Computer Interaction
- CNIT 31500 Systems Programming Spring 2024  
C Programming, data Structures and algorithms
- CNIT 35500 Software Development For Mobile Computers Fall 2023  
Android programming in Java
- CNIT 48300 Applied Machine Learning Fall 2023  
Applied Machine Learning in Tensorflow
- CNIT 39200 Enterprise Data Management Spring 2022  
Advanced database techniques (ERD, EER, UML) and Oracle SQL and NoSQL
- CNIT 27200 Database Fundamental Fall 2020 – Fall 2021, Fall 2022  
Entity-Relationship Diagrams (ERD) and Structured Query Language (SQL)

## INVITED TALKS & WORKSHOPS

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**Guest Lecturer**, University of Tennessee, Knoxville, TN *July 2026*  
Title: AI Digital Storytelling for Immigrant Mental Health From Pilot Insights to Practice  
SOWK 550, University of Tennessee, Knoxville

**Speaker**, University of Tennessee, Knoxville, TN *March 2026*  
Title: AI Storytelling for Immigrant Mental Health: From Pilot to Practice  
eVOLve: The Future of Behavioral Health Conference, College of Social Work Continuing Education Program

**Guest Lecturer**, Purdue University, West Lafayette, IN *June 2025*  
Purdue-UTK Community-based Mental Health Workshop, AI for Mental Health Support: designed and deployed a web-based conversational AI system & conducted a pilot study

**Guest Lecturer**, 4-H Academy Workshop, Purdue University, West Lafayette, IN *June 2025*  
Interactive AI: led a workshop, Blockly activities, and facilitated discussions on social aspects of AI

**Guest Lecturer**, Pai Chai University, Daejeon, South Korea *June 2022*  
IT Graduate School Training Program, Affective Robotics

## PUBLICATION

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### Under Review

1. anonymous authors, title omitted for anonymity, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); accepted
2. anonymous authors, title omitted for anonymity, Journal of Technology in Human Services
3. anonymous authors, title omitted for anonymity, Society for Social Work and Research (SSWR)
4. anonymous authors, title omitted for anonymity, Society for Social Work and Research (SSWR)

### Journals

- [J3] Wonse Jo\*, Ruiqi Wang\*, **Go-Eum Cha**, Su Sun, Revanth Senthilkumaran, Daniel Foti, and Byung-Cheol Min (\* equal contribution), MOCAS: A Multimodal Dataset for Objective Cognitive Workload Assessment on Simultaneous Tasks. IEEE Transactions on Affective Computing, 2024. DOI: 10.1109/TAFFC.2024.3414330

- [J2] Wonse Jo, **Go-Eum Cha**, Daniel Foti, and Byung-Cheol Min, SMART-TeleLoad: A New Graphic User Interface to Generate Affective Loads for Teleoperation. *SoftwareX*, 26, pp.101757, 2024. DOI: 10.1016/j.softx.2024.101757.
- [J1] **Go-Eum Cha**, Doeun Kim and Kang Hee Lee, System design of symptom detection of mild cognitive impairment with observational robot. *Journal of Theoretical and Applied Information Technology*, 97(18), pp.4824-4837, 2019. ISSN:1992-8645.

### Proceedings

- [C7] Yanfu Liu, **Go-Eum Cha**, Clayton Xu, Wenyan Wang, Ivy Leung, Tracy Kwan, Laura Tom, Maja Jacobs, Betina Yanez, Melissa Simon, Grace Chan, & Sooyeon Jeong. (2026). Designing a Social Robot to Enhance Digital Health Literacy in Cantonese-speaking Chinese Immigrant Communities. In *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems* (pp. 1-6). DOI: 10.1145/3772363.3798622
- [C6] Hieu Tran, **Go-Eum Cha**, & Sooyeon Jeong (2025). A Robot That Listens: Enhancing Self-Disclosure and Engagement Through Sentiment-based Backchannels and Active Listening. In *2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)* (pp. 1741-1748). IEEE. DOI: 10.1109/RO-MAN63969.2025.11217862
- [C5] Hyung Chan Cho, **Go-Eum Cha**, Yanfu Liu & Sooyeon Jeong (2025). Motivating Students' Self-study with Goal Reminder and Emotional Support. In *2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)* (pp. 1670-1677). IEEE. DOI: 10.1109/RO-MAN63969.2025.11217822
- [C4] **Go-Eum Cha**, Wonse Jo, & Byung-Cheol Min (2023). Implications of Personality on Cognitive Workload, Affect, and Task Performance in Remote Robot Control. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2023)*, Detroit, USA, October 1-5, 2023. DOI: 10.1109/IROS55552.2023.10341633
- [C3] **Go-Eum Cha**, & Byung-Cheol Min (2022). Correlation between Unconscious Mouse Actions and Human Cognitive Workload. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts* (pp. 1-7). DOI: 10.1145/3491101.3519658
- [C2] Wonse Jo, Shyam Sundar Kannan, **Go-Eum Cha**, Ahreum Lee, & Byung-Cheol Min (2020). ROSbag-based Multimodal Affective Dataset for Emotional and Cognitive States. In *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (pp. 226-233). IEEE. DOI: 10.1109/SMC42975.2020.9283320
- [C1] **Go-Eum Cha**, Eun Jin Kim, & Kanghee Lee (2017). State-of-the-Art of Domestic Emotional Robots. In *Proceedings of the Korean Society of Computer Information Conference* (pp. 161-162). Korean Society of Computer Information.

### Preprints & Posters

- [P8] Dayoon Suh, **Go-Eum Cha**, & Byung-Cheol Min (2024). Activity-based Behavioral Cooking Proficiency Estimation with Graph Neural Networks and Transformer, Spring Undergraduate Research Conference, Purdue University, USA.
- [P7] Soomin Kim, **Go-Eum Cha**, & Byung-Cheol Min (2023). Creating a safer future: The need for datasets on intrusive human behavior towards service robots, Spring Undergraduate Research Conference, Purdue University, USA.
- [P6] **Go-Eum Cha**, & Byung-Cheol Min (2023). Operator Personality and Task Performance in Robot Surveillance: A Statistical Analysis, Realizing the Digital Enterprise, Polytechnic Institute, Purdue University.

- [P5] Soomin Kim, **Go-Eum Cha**, & Byung-Cheol Min (2022). A Mobile Application for Customizable Robot Facial Interface, Purdue Polytechnic Undergraduate Research Expo, Polytechnic Institute, Purdue University.
- [P4] **Go-Eum Cha**, & Byung-Cheol Min (2021). Outer Space Rover Localization through Prior Object Information, Realizing the Digital Enterprise, Polytechnic Institute, Purdue University.
- [P3] **Go-Eum Cha**, Wonse Jo, & Byung-Cheol Min (2020). Human Cognitive Load Prediction with Behavioral Cues for Human-Machine Interaction, Realizing the Digital Enterprise Mini-Talk Session, Polytechnic Institute, Purdue University. [**Best Presentation Award**]
- [P2] Wonse Jo, Shyam Sundar Kannan, **Go-Eum Cha**, Ahreum Lee, and Byung-Cheol Min (2020). A ROS-based Framework for Monitoring Human and Robot Conditions in a Human-Multi-robot Team. *arXiv preprint arXiv:2006.03784*
- [P1] **Go-Eum Cha**, Seongha Park, & Eric Matson (2019). Autonomous Unmanned Aerial Vehicle Control System using 2.4GHz Radar. In 2019 IEEE Sensors Applications Symposium, March 11-13, France

## PROFESSIONAL ACTIVITIES & EXTRACURRICULAR

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### Reviewer:

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|--------------------------------------------------------------------------------------------|------------|
| ACM CHI conference on Human Factors in Computing Systems Extended Abstracts                | 2026       |
| ACM Conversational User Interfaces (CUI)                                                   | 2026       |
| IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)                 | 2025–2026  |
| ACM/IEEE International Conference on Human-Robot Interaction (HRI)                         | 2025       |
| ACM CHI conference on Human Factors in Computing Systems                                   | 2025       |
| IEEE International Conference on Robotics and Automation (ICRA)                            | 2025, 2021 |
| IEEE Open Journal of the Computer Society (OJCS)                                           | 2025       |
| IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)        | 2023–2024  |
| IEEE RAS EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob) | 2024       |
| IEEE International Conference on Intelligent Autonomous Systems (CIAS)                     | 2023       |
| International Journal of Fuzzy Logic and Intelligent Systems (IJFIS)                       | 2021       |
| IEEE Transactions on Human-Machine Systems (THMS)                                          | 2021       |

### Special Recognitions:

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|----------------------------------------------------------|------|
| Special Recognitions for Outstanding Reviews, ACM CHI EA | 2026 |
| Special Recognition for Outstanding Reviews, ACM CUI     | 2026 |

### Competitions:

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| NASA Space Robotics Challenge Phase 2 (Purdue-Hongik Team)                                                                                                                                      | 2021 |
| Autonomous multi-robot systems for exploration, resource localization, and return-to-base coordination in unknown lunar environments using Gazebo, SLAM, navigation, and YOLO-based perception. |      |

### Service:

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|-------------------------------------------------------------------------------------------------------------|------|
| Student Judge, Spring Undergraduate Research Conference, Purdue University, USA                             | 2023 |
| Student Judge, IEEE Sensors and Measurement Student Contest, 2019 IEEE Sensors Applications Symposium (SAS) | 2019 |

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**GRANTS**

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|-------------------------------------------------------------|-------------|
| IEEE/RSJ IROS Student Travel Grant                          | <i>2023</i> |
| Gary Marsden Student Development Fund Travel Award, ACM CHI | <i>2022</i> |

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**SELECTED AWARDS & HONORS**

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|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>2023 IEEE/RSJ IROS 2023 Travel Grant Awards</b>                                                                                      | <i>2023</i>                     |
| IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2023), USA                                                    |                                 |
| <b>Daniel &amp; Martina Lewis Grad Scholarship</b>                                                                                      | <i>2020 - 2021, 2022 – 2023</i> |
| Purdue University, USA                                                                                                                  |                                 |
| <b>Gary Marsden Travel Awards</b>                                                                                                       | <i>2022</i>                     |
| for attending ACM CHI 22', ACM SIGCHI, USA                                                                                              |                                 |
| <b>Best Presentation Award</b>                                                                                                          | <i>Dec. 2020</i>                |
| Realizing the Digital Enterprise (RDE) Student Showcase “Mini-Talk”<br>Polytechnic Institute, Purdue University, West Lafayette, IN USA |                                 |
| <b>Graduate Student Travel Grant</b>                                                                                                    | <i>Mar. 2019</i>                |
| Department of Computer Information and Technology, Purdue University, USA                                                               |                                 |
| <b>Office of Globalization International Travel Support Funds</b>                                                                       | <i>Mar. 2019</i>                |
| Purdue Polytechnic Institute, Purdue University, USA                                                                                    |                                 |